Insecurity of Lost/Stolen Phone Reporting Services: Vulnerabilities, Attacks, and Countermeasures

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Abstract

Lost and stolen phone reporting services are widely deployed to prevent unauthorized device use by blacklisting International Mobile Equipment Identities (IMEIs). However, through an extensive experimental study across three major U.S. carriers and diverse mobile devices, we discover that these services unexpectedly introduce severe and previously unexplored security risks. Specifically, we identify six new vulnerabilities spanning the device, carrier, and cross-carrier domains, which together enable attackers to arbitrarily block cellular devices from accessing carrier networks. Building on these findings, we design and validate two practical denial-of-service (DoS) attacks: Home Security System Freezing, which disables cellular-based home security gateways and blocks alarm delivery, and Zero-Day Flagship Phone Ambush, which preemptively blocks brand-new flagship phones from accessing mobile services at launch. Both attacks are experimentally validated on operational 5G/4G networks. Finally, we propose practical, backward-compatible countermeasures and implement a prototype to evaluate their effectiveness.

CCS Concepts

• **Security and privacy → Mobile and wireless security**.

Keywords

Cellular Networks, Security, Lost Phone Reporting Services

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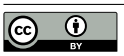
1 Introduction

The problem of stolen or lost mobile phones has grown increasingly severe in recent years. A 2025 GSMA (GSM Association) report [42] estimates that about 1% of active subscriptions each year involve stolen phones, totaling millions of devices and $10.8 billion in annual losses. Stolen devices can be repurposed for phishing, DDoS, and misinformation attacks, posing significant risks to the mobile ecosystem. This risk is mitigated in practice by the 3GPP (3rd Generation Partnership Project)-defined Equipment Identity Register (EIR) [11], which supports lost and stolen phone reporting services. Mobile operators maintain their own EIRs, which are local databases that manage devices' International Mobile Equipment Identities (IMEIs) [8] within the operator's network. The EIR is responsible for enforcing device-level access control by classifying IMEIs into different lists (e.g., whitelist and blacklist), thereby determining whether a device is allowed to access network services. In addition to local EIRs, the Central EIR (CEIR), deployed by GSMA [39], serves as a global coordination platform that aggregates IMEI records contributed by participating operators. The CEIR enables cross-operator synchronization of device status by distributing blacklisted IMEIs, allowing operators worldwide to consistently block devices reported as lost or stolen.

During device registration [3, 9], the network obtains the device's IMEI and queries the EIR. If the IMEI is blacklisted, registration is rejected and network access is denied. Today, over 800 operators worldwide support this service, allowing users to blacklist IMEIs. In practice, users report lost or stolen phones through three main channels: (1) in-store visits, (2) phone calls or online chats, and (3) online self-reporting portals.

However, this service is vulnerable to abuse due to the lack of formal ownership verification, unlike government-registered properties such as houses or vehicles. IMEIs are often absent from purchase receipts, making ownership difficult to prove. When IMEIs can be spoofed, the reporting service becomes exploitable. Although IMEI-spoofing defenses [17, 19, 30, 34, 49, 51, 59, 64] have been studied for over a decade, none have achieved global deployment due to hardware, legacy-network, and cross-generation compatibility constraints. Carriers and 3GPP thus introduced several pragmatic safeguards for the reporting services. However, these safeguards remain largely unexplored, motivating us to examine these security defenses for reporting services.

Three major defenses are commonly adopted. First, anti-abuse reporting requirements restrict report submissions to subscribed users only, introducing traceability and increasing attack cost. Second, IMEI protection is mandated by 3GPP; devices must not transmit their IMEIs in plaintext before Non-Access Stratum (NAS) security is activated [5, 6], preventing over-the-air IMEI leakage. Third, possession-based ownership is partially verified in practice that carriers rely on device activity patterns, such as historical usage or subscription association, rather than solely on the IMEI. Collectively, these defenses are expected to raise the cost and significantly reduce adversaries' incentives of abusing lost and stolen phone reporting services.



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Category	Vulnerability	Description	Empirical Validation		Proof-of-Concept Attacks
			Carriers	Devices	
Incomplete IMEI Protection on Mobile Devices (§4)	V1: Weak Integrity Check on Identity Request Messages (§4.1)	Due to weak integrity checks, some devices may incorrectly disclose their IMEIs when receiving malformed requests.	OP-I, OP-Ia†, OP-Ib†, OP-II‡, OP-IIa†, OP-IIb†, OP-III, OP-IIIa†, OP-IIIb†	Samsung S21, iPhone SE3, iPhone 16 Pro, Samsung Galaxy Z Fold 7, Sixfab CIoT HAT, Botletics SIM7000, Telit Charlie Kit, MKR NB 1500, HA-I, HA-II*	**Home Security System Freezing**♯✠**:** an innovative DoS attack leveraging V1, V2, V3, and V4 incapacitate home security systems and prevent both homeowners and security service centers from receiving critical alarms (§7). **Zero-Day Flagship Phone Ambush**✠**:** a zero-day attack exploiting V2, V3, V6 to cause hundreds of flagship phones (e.g., Samsung Galaxy Z Fold 7) to be blocked from mobile services immediately upon debut (§8).
Inadequate Reporting Security on Carrier Infrastructure (§5)	V2: Required but Absent Reported Identity Verification (§5.1)	Individuals without ID verification, or with only minimal verification, can still submit lost/stolen reports.			
	V3: Insufficient Usage-based Device Ownership Verification (§5.2)	Without standards guiding usage configuration, usage-based checks can be set too weak, letting adversaries fake activity and report devices they don't own.			
	V4: Reporting Devices Beyond Service Subscription Entitlements (§5.3)	Reporting systems accept arbitrary IMEIs without verifying whether the device is covered by the subscriber's service plan.			
Loose Governance on Cross-Carrier Operations (§6)	V5: Non-Mandatory CEIR Participation (§6.1)	GSMA CEIR participation is not mandatory, allowing adversaries to bypass global device blocking.			
	V6: Missing Verification Requirements During CEIR/EIR Synchronization (§6.2)	Downloaded device blocking records lack security metadata—such as reporter identity verification status—needed to detect dubious or malicious reports.			

†: Associated MVNOs were used exclusively to test V6.
‡: V6 was not observed in OP-II or in its associated MVNOs.
♯: Our validation was conducted on OP-II, as it is the default carrier used by HA-I.
*: HA-II is observed to use the same vulnerable cellular modem as HA-I.
✠: As this study focuses on the insecurity of lost/stolen phone reporting services, the devised attacks share common vulnerabilities, specifically V2 and V3. However, the required vulnerabilities vary across attacks, and not all vulnerabilities are necessary in every attack. For example, the Zero-Day Flagship Phone Ambush attack additionally requires V6, whereas the Home Security System Freezing attack requires V1 and V4.

Table 1: Summary of vulnerabilities and proof-of-concept attacks in this work.

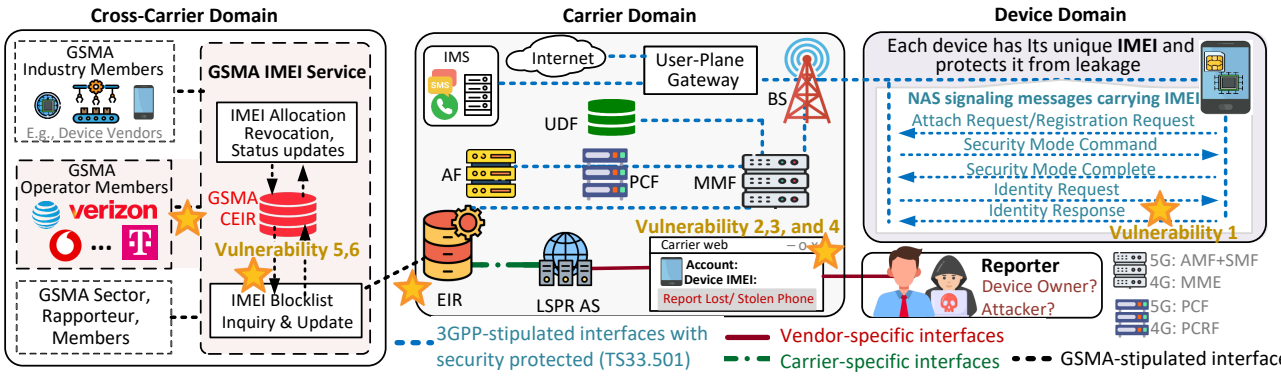


Figure 1: End-to-end lost/stolen device reporting ecosystem across device, carrier, and cross-carrier domains, with vulnerabilities (V1–V6) shown at their exposed points.

Surprisingly, our experimental study across three major U.S. carriers using diverse mobile devices still reveals that the deployed security mechanisms are insufficient. Table 1 summarizes our findings. We identify six new vulnerabilities spanning the device, carrier, and cross-carrier domains, which fall into three major vulnerability categories: incomplete IMEI protection on mobile devices, inadequate reporting security on carrier infrastructure, and loose governance on cross-carrier operations. The root causes are unexpectedly complex and stem from the interplay of standards, carrier implementations, and cross-carrier operations.

By exploiting these vulnerabilities, we develop two proof-of-concept attacks: (1) Home Security System Freezing (secFreezing), which disables critical home security infrastructure; and (2) Zero-Day Flagship Phone Ambush (flagAmbush), which blocks brand-new, unboxed flagship phones such as the Samsung Z Fold 7 from accessing mobile services immediately upon release. Our experimental results show that victims receive no notification from their operators and silently lose cellular connectivity when their IMEIs are maliciously reported as lost or stolen. Although victims can contact their operators to request service restoration, the process is time-consuming, as it typically requires proof of ownership (e.g., purchase receipts containing the IMEI) and identity verification. Furthermore, if the victim's carrier differs from that of the reporting party, the request may need to be escalated to the GSMA for additional review, introducing further delays. This disruption window creates an opportunity for adversaries to carry out attacks, such as facilitating home intrusions by disabling security systems.

Moreover, launching the proposed attacks incurs low cost: an entry-level Software-Defined Radio (SDR) platform (e.g., USRP B210), legacy smartphones (e.g., LG G3), and a small number of mobile user accounts suffice. The potential benefits for adversaries can be substantial. For example, the secFreezing attack enables adversaries to disrupt alarm delivery and gain unauthorized access to homes without detection, leading to financial losses for homeowners. The flagAmbush attack can prevent newly released devices from accessing cellular networks at debut, causing significant reputational and market damage to manufacturers. Adversaries could exploit such disruptions for financial gain, for instance, by short selling the affected company's stock prior to launching the attack.

Finally, we propose practical solutions and evaluate their effectiveness on a prototype platform. Our findings have been responsibly disclosed to affected parties, including carriers, chipset vendors, and device manufacturers. GSMA has also acknowledged the significance of these security issues and the proposed remedies.

2 Background

In this section, we provide an overview of the 5G/4G mobile network and device ecosystem, along with the cross-carrier IMEI service. We also summarize current threats to lost and stolen reporting services and the corresponding defenses.

2.1 5G/4G Mobile Network And Device

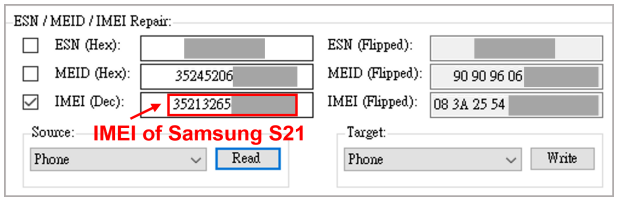
Figure 1 illustrates the carrier domain, which shows the architecture of a 5G/4G network that supports data, voice, text, and lost/stolen phone reporting services. When the User Equipment (UE) initiates a service request, the request passes through the Base Station (BS) and the core network, and then reaches either the IP Multimedia Subsystem (IMS) for multimedia services or the Internet for data services. The UE includes both non-IoT devices, such as mobile phones, and IoT devices. The BS provides these devices with wireless access to the core network.

The core network has two main planes. The user plane forwards user traffic to the IMS or the Internet through the User Plane Gateway (UPG), which provides IP connectivity. The control plane consists of six key functions: (1) the Mobility Management Function (MMF), which handles mobility management, authentication, resource reservation, and IP connectivity setup; (2) the User Data Function (UDF), which stores user data and subscription information; (3) the Policy Control Function (PCF), which generates billing rules, QoS parameters, and routing policies; (4) the EIR, which manages device status (e.g., whitelist/blacklist) based on IMEI and provides interfaces, such as PDBI for Oracle EIR [63], for adding or removing device entries; (5) the Application Function (AF), which is allowed to deploy custom service applications; and (6) the LSPR AS (Lost/Stolen Reporting Service Application Server), which is a carrier-deployed server for user reporting of lost/stolen devices and updating the EIR.

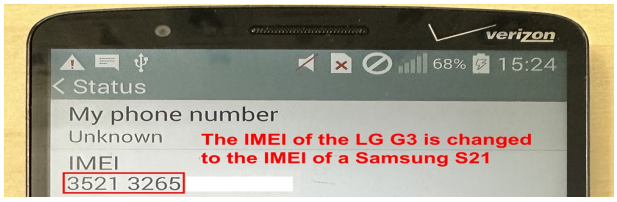
Device Identity Management. The IMEI is a globally unique 15-digit identifier assigned to devices with 3GPP transmitters [32], independent of users, SIM cards, and service subscriptions, such as voice and data. Managed and allocated by the GSMA, IMEIs are embedded by manufacturers during production, ensuring uniqueness and traceability. When a user connects their device to a cellular network using their service subscription, its IMEI is recorded by the serving operator, allowing operators to maintain subscriber–IMEI mappings. In practice, through carrier portals, users can find the IMEIs of all devices—including data-only devices—associated with their accounts, including both individual and family accounts. Therefore, users can retrieve the IMEI of a lost device through the carrier portal or customer support to access the lost/stolen reporting service.

Device Status Verification. When a device attempts network registration, the UE sends a Registration Request to the MMF, followed by mutual authentication. After successful authentication and the establishment of NAS and AS security channels [7], the MMF issues an Identity Request, and the UE returns its IMEI. The MMF then checks the IMEI with the EIR: if the device is blacklisted, registration is rejected; otherwise, registration continues.

Lost/Stolen Phone Reporting Services. Most carriers offer three channels for submitting lost or stolen phone reports: (1) in-store visits, (2) phone calls or online chats, and (3) online self-reporting portals. For the first two options, users must supply the IMEI of the lost device and may be required to complete additional ownership or identity verification, such as presenting a purchase receipt, SSN, or driver's license information. In contrast, the online self-reporting option automatically selects the IMEI associated with the user's account and requires no further ownership verification.



(a) EFS Tool: Replacing the IMEI of an LG G3 with the IMEI of a Samsung S21.



(b) LG G3' IMEI has been replaced with the IMEI of the Samsung S21.

Figure 2: IMEI spoofing using the EFS tool [1].

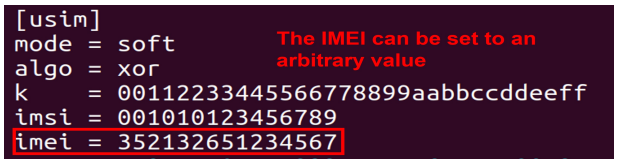


Figure 3: IMEI spoofing using an SDR-based UE.

2.2 Cross-Carrier IMEI Service

The GSMA IMEI service provides global device-identity management through two main functions: (1) maintaining the IMEI database (IMEI DB), which tracks IMEI allocations, revocations, and status updates, and (2) authorizing operators and industry members/partners to query it for IMEI validation, device identification, and status checks.

Centralized Equipment Identity Register (CEIR). The CEIR is a centralized repository built atop the IMEI DB that aggregates operators' EIR updates, maintains global white/black/grey lists, and distributes a unified block list. A device blacklisted by one operator is therefore blocked across all CEIR-connected networks. Carriers upload block-list updates to the CEIR hourly, and others retrieve and apply them on the same schedule [38, 40].

2.3 Current Threats and Defenses

Threats. IMEI spoofing is the primary threat to lost or stolen phone reporting services. By rewriting a device's IMEI, attackers can impersonate other devices, connect the device to cellular networks, and abuse carrier services or policies, such as obtaining cheaper data plans [94]. In practice, there are two approaches for adversaries to spoof the IMEIs of other users' devices: (1) using commercial mobile phones and (2) using Software-Defined Radio (SDR) UE platforms. In the former approach, adversaries modify the IMEI stored in a device's modem non-volatile (NV) memory, where the IMEI is stored, using tools such as EFS Professional [1]. For example, the IMEI of an LG G3 can be replaced with that of another device

(e.g., Samsung S21) using the EFS tool, as shown in Figure 2. The latter approach involves deploying an SDR-based UE platform with a configurable IMEI, such as srsRAN [77], as illustrated in Figure 3.

Defenses against IMEI Spoofing. Existing defenses against IMEI spoofing mainly fall into three categories: (1) *Software-based methods* (e.g., [17, 19, 30, 34, 51]) detect abnormal IMEI changes via system log monitoring [17, 30, 51] or checksum validation (e.g., the Luhn algorithm) [19, 34]. However, as these approaches rely on on-device software enforcement, they can be bypassed by SDR-based UE platforms that do not implement such checks. (2) *Hardware-based methods* (e.g., [49, 59]) bind IMEIs to secure hardware (e.g., TrustZone [18] and Trusted Execution Environments (TEEs) [93]), preventing tampering. However, they require specialized hardware, limiting deployment on resource-constrained devices. (3) *Infrastructure-based methods* enable operators to verify whether the IMEIs transmitted by devices legitimately belong to them. For example, the IMEI authentication [48] requires a device to digitally sign a statement containing its IMEI, allowing operators to validate it using device certificates (e.g., X.509 [75]). However, these approaches often require updates to 3GPP standards (e.g., to support IMEI signature verification), making them impractical for legacy networks with frozen standards.

Operational Defenses in Reporting Services. Although IMEI-spoofing defenses have been proposed for over a decade, none have achieved global deployment due to hardware, legacy-network, and cross-generation compatibility constraints. As a result, carriers and 3GPP standards bodies introduced pragmatic safeguards—not to prevent spoofing outright, but to limit its impact and reduce the incentives for abusing lost/stolen reporting services. (1) *Anti-abuse reporting requirement:* only subscribed users may file lost/stolen reports, adding traceability and raising attack cost. (2) *Device IMEI protection:* 3GPP requires that IMEIs not be sent in plaintext before NAS security activation [4, 5], reducing over-the-air IMEI exposure. (3) *Possession-based ownership checks:* carriers validate ownership through device activity rather than relying solely on the IMEI.

3 Unexplored Attack Surfaces in Lost/Stolen Phone Reporting

In this work, we investigate the end-to-end lost/stolen phone reporting services in mobile networks across three critical domains: device, carrier, and cross-carrier. Following this overview, we present our threat model, methodology, and ethical considerations.

⋄*Device domain.* For reporting services that rely on IMEI-based identification, the key attack surface in the device domain lies in how the IMEI is protected and delivered. Even when victim UEs are uncompromised, non-standard implementations introduced for legacy compatibility or incomplete security enforcement by device vendors during IMEI delivery can cause the IMEI to be exposed before authentication and before any security context is established. We analyze this practical attack surface in § 4.

⋄*Carrier domain.* This domain's attack surface concerns whether a received lost or stolen report is properly authorized, which depends primarily on verifying the reporter's identity. We examine carriers' identity-verification mechanisms in practice in § 5.1 and 5.2. Furthermore, even if identity verification fails, the damage can be limited if the carrier enforces restrictions tied to the reporter's subscription. We evaluate whether such restrictions exist in § 5.3.

⋄*Cross-Carrier domain.* Cross-carrier device blacklisting is specified by GSMA [38, 40] but lacks strict regulatory requirements, leaving its implementation across carriers as a potential attack surface. Carriers adopt heterogeneous reporting workflows, resulting in inconsistent security practices. This heterogeneity is further amplified by the coexistence of Mobile Network Operators (MNOs) and Mobile Virtual Network Operators (MVNOs). If some carriers do not participate, lost or stolen devices can still operate on non-participating networks. Conversely, among participating carriers, weaknesses in one carrier's reporting or verification mechanisms can propagate across others, amplifying the overall impact. We analyze vulnerabilities in this domain in § 6.

Threat Model. Victims are owners or vendors of devices with 3GPP transceivers (e.g., phones, wearables, home-security gateways, vending machines). Neither the cellular networks nor the victim UEs are compromised in any attack. Adversaries are individuals or organizations launching device-level DoS attacks against victims. They: (1) can access the carrier-provided reporting services available to normal users and may inject messages over public wireless channels, but cannot break cryptographic protections (e.g., cannot obtain ciphering or integrity-protection keys); (2) can subscribe to cellular services and attach their devices to cellular networks; and (3) exploit only the vulnerabilities identified in this work, assuming no additional weaknesses in the networks or devices.

Experimental Methodology. We validate the vulnerabilities on operational networks of three major U.S. carriers (OP-I, OP-II, OP-III) and their dependent MVNOs (OP-Ia/b, OP-IIa/b, OP-IIIa/b). Our experiments involve five types of UEs: (1) SDR-based UEs built on srsRAN [77]; (2) commercial off-the-shelf (COTS) smartphones, including the LG G3, Apple iPhone SE3 and 16 Pro, and Samsung Galaxy S21 and Z Fold 7; (3) wearable devices, such as the LG Watch Urbane; (4) cellular IoT modem kits, including the Sixfab CIoT Hat [76], Botletics SIM7000 [27], and the Telit Charlie kit [84]; and (5) home security gateways from two representative U.S. alarm system providers, denoted HA-I and HA-II.

Ethical Consideration and Responsible Disclosure. We recognize that certain feasibility tests and attack evaluations may pose risks to cellular carriers and users. To conduct this study responsibly, we adopt three measures. First, we use only our own devices as victims and purchase unlimited service plans from all tested carriers. Second, all vulnerability validation and attack experiments are performed at a small scale, strictly following the principle of identifying security issues without causing harm. Third, we submit our research methodology to our institute's IRB for review and receive a waiver, as no human subjects are involved.

We have reported our findings to all affected carriers, chipset vendors, device manufacturers, and relevant international standards organizations (e.g., GSMA), along with recommended mitigations. Carriers and vendors are still investigating the reported vulnerabilities. GSMA, however, has acknowledged the significance of our work, and its security manager has forwarded our findings to the device security group for further discussion and remediation.

4 Incomplete IMEI Protection on Mobile Devices

3GPP standards require UEs to withhold their IMEIs until NAS security is activated [4, 5], but this requirement is not always enforced

Bits	Field	Octet
8–5	Security header type	1
4–1	Protocol discriminator	
Message authentication code		2–5
Sequence number		6
NAS message		7–n

Table 2: General message organization example for a security protected NAS message[4, 5].

in practice because current 3GPP UE conformance tests [2, 15] do not examine IMEI handling under malformed NAS procedures. To assess whether IMEI leakage occurs, we analyze all six 5G/4G NAS messages capable of triggering or carrying IMEI transmission: Identity Request, Identity Response, Security Mode Command, Security Mode Complete, Attach Request, and Registration Request. Our results show that not all devices implement IMEI protection as specified, leaving them vulnerable to IMEI leakage.

4.1 V1: Weak Integrity Check on Identity Request Messages

As specified in 3GPP standards [4, 5], NAS messages carry a Security Header Type field that specifies their protection mode, as shown in Table 2. This field indicates whether a message is plaintext, integrity-protected, or integrity-protected-and-ciphered. A UE is required to verify that the indicated mode matches the actual protection state and to discard any message with an invalid MAC or missing encryption.

In this work, we examine whether devices correctly enforce these checks across all the aforementioned three protection modes. Specifically, we test whether Identity Request and Security Mode Command messages can still be processed, and IMEIs disclosed, when they are malformed (e.g., with an incorrect MAC or an unencrypted payload despite the header indicating protection). Acceptance of such messages would allow adversaries to harvest victims' IMEIs and launch IMEI-based attacks.

Experimental Validation. To evaluate this vulnerability, we tested four commercial smartphones (Google Pixel 5, iPhone SE3, Samsung Galaxy S8 and S21) and four IoT development boards (Sixfab CIoT HAT [76], Botletics SIM7000 [27], Telit Charlie (ME310G1) [84], and MKR NB 1500 (SARA-R410M) [20]). We crafted Identity Request and Security Mode Command messages under different protection modes and sent them to each device to observe whether the device responded by returning its IMEI.

Our experiments reveal two results. First, all devices correctly ignored malformed requests using the plaintext or integrity-protected-and-ciphered modes, as required by 3GPP. Second, two IoT boards, Sixfab CIoT HAT and Botletics SIM7000, incorrectly disclosed their IMEIs when receiving requests in the integrity-protected mode with an invalid MAC (Figure 4), whereas the other devices did not.

Note that although only two cellular IoT boards exhibited this behavior, the issue lies in the underlying modem chipsets rather than the platforms themselves. Because NAS security is implemented in the cellular modem, any device using these chipsets is likely vulnerable. These chipsets come from two leading cellular IoT modem vendors whose combined global market share exceeds 40%,

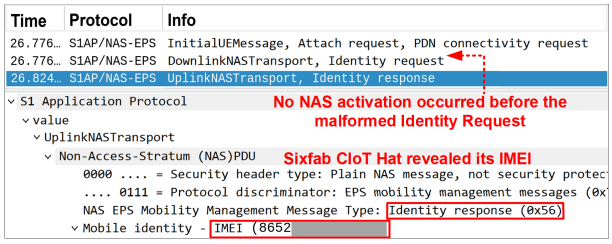


Figure 4: Sixfab CIoT Hat discloses its IMEI when NAS security is not activated yet.

suggesting that this weakness may affect a significant portion of the IoT ecosystem.

Root Cause and Lessons Learned. Although V1 initially appears attributable to IoT modem vendors whose devices disclose IMEIs when receiving unauthenticated NAS messages, the root cause is systemic. The 3GPP standards [4, 5] require IMEI protection, but the 3GPP UE conformance tests do not cover unauthenticated NAS procedures such as Identity Request and Security Mode Command. Because these procedures fall outside the 3GPP certification testing scope, chipset vendors can still obtain full certification despite violating these protections. This standards–testing gap has received little attention, despite its ecosystem-wide impact.

The key lesson is that IMEI protection cannot rely solely on vendor implementations; conformance tests must enforce the security guarantees already required by 3GPP. Without updating these tests to cover unauthenticated NAS procedures, vulnerable chipsets, including IoT modules, will continue to receive full certification, sustaining a scalable IMEI-harvesting attack surface.

5 Inadequate Reporting Security on Carrier Infrastructure

Lost and stolen device reporting is an essential security service provided by mobile carriers, yet its design is largely unregulated, as neither 3GPP nor GSMA specifies how such services should be implemented. As a result, carriers independently develop their reporting procedures, leading to diverse designs and potentially distinct security pitfalls.

In this work, rather than targeting carrier-specific flaws, we focus on security weaknesses that generalize across operators. From a user (reporter)-centric perspective, we examine the core requirements common to all designs, namely who is authorized to submit reports and which devices can be reported under existing subscription and ownership assumptions. By conducting this analysis using three major U.S. carriers, we uncover three recurring weaknesses: (V2) Required but Absent Reporter Identity Verification, (V3) Insufficient Usage-based Ownership Verification, and (V4) Reporting Devices Beyond Subscription Entitlements.

5.1 V2: Required but Absent Reporter Identity Verification

Our study of three major U.S. carriers shows that reporting channels are accessible only to subscribers with active plans, which should, in principle, ensure accountability by allowing carriers to trace malicious reports back to the responsible user. However, this assurance breaks down due to differences in subscriber account types. While postpaid accounts require strict identity verification, such

Subscription Plan & Mobile Carrier		Information Required for Account Setup								Lost/Stolen Phone Reporting
		Name	Date of Birth	Address	Email	Phone Number	Payment Info.	Social Security Number	Identity Document	
Post-paid	OP-I	●	●	●	●	●	●	●	●	All postpaid and prepaid users are allowed to use reporting services
	OP-II	●	●	●	●	●	●	●	●	
	OP-III	●	●	●	●	●	●	●	●	
Pre-paid	OP-I	◐	○	◐	◐	◐	●	○*	○*	
	OP-II	◐	○	◐	◐	○	●	○*	○*	
	OP-III	◐	○	◐	◐	○	●	○*	○*	

●: Information required and verified by the carrier ◐: Information required but not verified by the carrier
○: Information not required by the carrier
*: **None of the tested carriers requires SSN or ID verification to create a prepaid account for reporting lost/stolen phone.**

Table 3: Summary of the information required for prepaid and postpaid accounts by the tested carriers.

as SSN-based checks and government-issued IDs, prepaid accounts typically involve little or no verification. As a result, if prepaid users are allowed to submit loss reports without additional checks, minimally verified individuals can maliciously report devices as lost or stolen with little traceability.

Notably, as described in § 2.1, carriers can identify the IMEI associated with a user account, which may restrict adversaries to reporting only devices linked to their own prepaid accounts. However, these IMEI-to-account associations can be manipulated (as discussed in V3 (§ 5.2)), enabling adversaries to associate arbitrary devices with their accounts before submitting fraudulent reports.

Experimental Validation. We experimentally validated this vulnerability by examining each carrier's subscriber documentation and onboarding workflow, and then creating both postpaid and prepaid accounts. Table 3 summarizes our findings. Two observations consistently hold across all carriers. First, prepaid and postpaid users are both permitted to submit lost/stolen device reports. Second, none of the carriers verify SSNs or government-issued IDs when creating prepaid accounts.

This lack of verification from prepaid accounts is critical. Prior work shows that attackers can routinely register prepaid accounts using fabricated or stolen information [50, 85]. Although carriers collect payment details, these do not guarantee traceability, as anonymous Visa gift cards [90], widely available at retailers such as Walmart, function as credit cards but require no identification to purchase. As a result, unverified users can freely obtain prepaid accounts and directly access reporting interfaces, enabling anonymous device-level DoS attacks in practice.

Root Cause and Lessons Learned. Carriers relax identity checks for prepaid users to accommodate those without SSNs, such as international or short-term visitors. Although alternative government IDs (e.g., passports) exist, reporting services do not enforce a strong binding between the reporter and the submitted report. This structural gap, consistent across all tested carriers, allows minimally verified accounts to file lost or stolen reports. The key takeaway is that reporting services need explicit identity-binding mechanisms to ensure reports come from accountable users, regardless of subscription type.

5.2 V3: Insufficient Usage-based Device Ownership Verification

Carriers commonly rely on usage-based verification as a proxy for device ownership in lost or stolen reporting services, since no standardized ownership certificate exists across carriers or device generations. Consequently, a device's historical activity, such as call counts, data volume, or accumulated registration time, is used as evidence of ownership. However, neither 3GPP nor GSMA specifies how these metrics should be parameterized, leaving carriers to define thresholds independently. Because usage is only a heuristic and can be trivially fabricated by spoofed devices, weak thresholds allow adversaries to generate minimal activity and report devices they do not own.

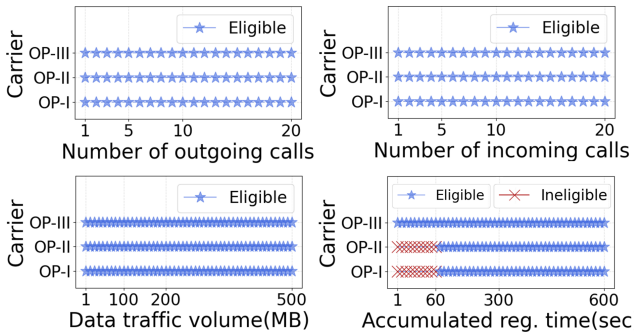


Figure 5: Carrier-specific device eligibility rules across calls, data usage, and registration time.

Experimental Validation. We evaluated four activity metrics across OP-I, OP-II, and OP-III: (1) incoming calls (1, 5, 10, 20), (2) outgoing calls (1, 5, 10, 20), (3) data usage (1 MB, 100 MB, 200 MB, 500 MB), and (4) accumulated registration time (1 s, 60 s, 300 s, 600 s). Our experiments used three devices: an SDR-based UE built on srsRAN [77] with a USRP B210, an LG G3 smartphone, and a Samsung Galaxy S21 as the victim device. The first two devices spoofed the S21's IMEI. Call-based tests used only the G3 because the SDR UE lacks voice capabilities.

The experiment proceeded as follows. First, we created a new prepaid account on each carrier; the reporter had never used the S21. Second, the attack devices spoofed the S21's IMEI (see § 2.3). Third, these devices attached to each network and generated activity according to the defined thresholds. Finally, we accessed each carrier's self-reporting portal to determine if the S21 became eligible for reporting as lost.

Figure 5 summarizes the results. We observed two key issues. First, registration-time requirements are extremely weak: OP-III accepts 1 s of attachment, and OP-I/OP-II require only 1 min. Second, once this minimal condition is met, carriers impose no additional

thresholds on calls or data usage. As a result, attackers can satisfy ownership checks with only brief attachment, enabling fully automated device-level DoS attacks at negligible cost. Exploiting V3 requires obtaining the IMEIs of target devices. This is feasible in practice; we present two methods for harvesting IMEIs of high-value targets in § 7 and § 8.

Root Cause and Lessons Learned. The fundamental limitation of usage-based verification is that usage does not imply ownership. Although IMEI spoofing defenses have been proposed for decades, none are deployable at scale due to practical constraints. For example, cryptographically binding IMEIs to public–private key pairs would prevent spoofing but is incompatible with resource-constrained IoT devices and would require modifications across multiple generations of 3GPP standards. Moreover, many legacy systems, especially 2G and 3G, have frozen specifications, making retrofit infeasible. Consequently, carriers rely on fragile usage heuristics as a backward-compatible proxy for ownership. A key lesson is that reliable ownership verification must incorporate multiple behavioral and contextual indicators rather than any single usage metric. Aligning long-term reporter–device interaction histories, such as recurring activity patterns, provides a robust and 3GPP-compatible path to strengthening reporting security.

5.3 V4: Reporting Devices Beyond Service Subscription Entitlements

Lost and stolen reporting services are intended for consumer smartphones, yet our analysis shows that carriers accept any IMEI a subscriber submits without verifying whether the device is covered by the subscriber's plan. Thus, users with only a standard smartphone plan can report devices provisioned under commercial or infrastructure IoT subscriptions [22, 89], including cellular home-security gateways, medical monitoring devices, and utility or industrial IoT modules. These systems operate under dedicated business IoT plans and support safety- and infrastructure-critical services.

Experimental Validation. To assess whether carriers enforce device-type or subscription-based restrictions, we evaluate two categories of devices on OP-I, OP-II, and OP-III: (1) *non-phone devices*, including wearables (e.g., LG Watch Urbane) and cellular IoT modem kits (e.g., Sixfab CIoT Hat [76], Botletics SIM7000 [27]), commonly used in commercial and infrastructure deployments; and (2) *unsupported phones* (e.g., Xiaomi Redmi) that cannot operate on the carriers' radio technologies or frequency bands. These serve as a sanity check: if even non-operational devices can be reported, then no device-type or validity constraints are enforced.

The experiment proceeds as follows. Using an SDR-based UE, we spoof each target device's IMEI, attach to each carrier's network using only a standard smartphone subscription, and detach. No wearable or IoT plans are purchased. We then submit lost/stolen reports for these IMEIs through each carrier's self-reporting portal.

Our results yield three findings. First, all non-phone devices and unsupported phones can be successfully reported across all carriers, and once reported, they are unable to reattach. Second, carriers inconsistently recognize IoT modem kits: OP-III misclassifies the Sixfab CIoT Hat as a mobile hotspot and leaves the Botletics SIM7000 unrecognized, yet both reports are still accepted. Third, carriers permit reporting of devices entirely incompatible with their own network technologies, confirming that no entitlement, device-type, or compatibility checks are applied.

In summary, users with ordinary smartphone plans can report wearables, IoT modules, unrecognized devices, and even devices incompatible with the network—indicating that reporting services impose no plan- or device-based restrictions.

Root Cause and Lessons Learned. The root cause of this vulnerability is that carriers perform neither subscription-entitlement checks nor basic device validation. Reporting systems accept arbitrary IMEIs without verifying whether the device is covered by the subscriber's plan, belongs to an appropriate device class, or is even operational on the network. This enables consumer subscribers to disable devices far outside their service scope, including commercial and infrastructure IoT systems.

A key lesson is that reporting services must enforce both entitlement checks and device-type validation, ensuring that only devices legitimately associated with a subscriber's plan—and technically compatible with the network—can be reported. Such validation is lightweight, backward-compatible, and essential to preventing cross-category abuse.

6 Loose Governance on Cross-Carrier Operations

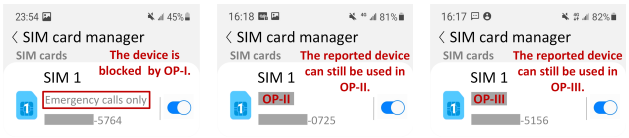
Cross-carrier device blocking is coordinated between the GSMA and Contributor Network Operators (CNOs)—typically MNOs such as AT&T and T-Mobile—via the CEIR using GSMA-stipulated protocols and policies [38, 40]. CNOs must synchronize their local EIR block lists with the CEIR hourly and block all downloaded IMEIs. Similar synchronization also occurs between CNOs and their MVNOs (e.g., Tello) through vendor-specific EIR interfaces [28, 60, 63, 87]. Under FS.45 [38], CNOs must provide MVNOs with access to the CEIR, either directly or as delegated contributors.

In principle, this ecosystem should prevent unauthorized use of lost or stolen devices, but loose governance undermines its effectiveness. Our study reveals two structural issues: (1) CEIR participation is optional even for GSMA operator members (V5), enabling adversaries to bypass global blocking; and (2) CEIR/EIR data exchanges lack mandatory security metadata—such as reporter-identity verification status—resulting in IMEI pollution across carriers (V6).

6.1 V5: Non-Mandatory CEIR Participation

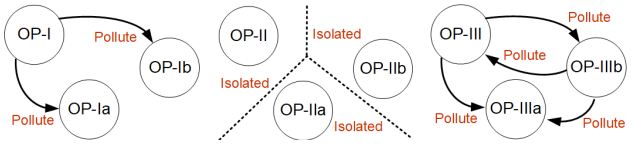
OP-I, OP-II, and OP-III—the three major U.S. carriers and GSMA operator members—collectively serve over 300 million subscribers [92]. Given their scale and their public claims of sharing lost/stolen device information with other carriers and international databases [23, 82], they should be natural CEIR participants. However, our findings show otherwise: a device blocked by one carrier remains fully usable on the others, allowing adversaries to bypass global blocking.

Experimental Validation. To confirm this, we reported a device as lost to one carrier (e.g., OP-I) and waited one week—far exceeding the hourly CEIR synchronization interval [38]. We then tested whether the same device functioned on the remaining carriers (OP-II and OP-III). Our results show that a device reported as lost in OP-I remains fully usable in OP-II and OP-III (Figure 6), and vice



(a) OP-I. (b) OP-II. (c) OP-III.

Figure 6: Device reported as lost in OP-I, still functions in OP-II and OP-III.



(a) OP-I and its MVNOs. (b) OP-II and its MVNOs. (c) OP-III and its MVNOs.

Figure 7: EIR pollution varies with MNOs and MVNOs.

versa. This indicates that at least two major U.S. carriers do not participate in CEIR synchronization, contradicting their claims.

Root Cause and Lessons Learned. The root cause of V5 is that participation in the GSMA CEIR—responsible for cross-carrier IMEI blocklist synchronization—is not mandatory. FS.45 [38] defines CEIR usage only as a recommended practice, leaving operators free to omit updates, delay propagation, or ignore revocations. This non-mandatory design produces inconsistent IMEI states across carriers and enables attackers to exploit propagation gaps. A key lesson is that cross-carrier blocking requires CEIR participation to be elevated from optional guidance to a mandatory requirement for all operator members.

6.2 V6: Missing Verification Requirements During CEIR/EIR Synchronization

During CEIR/EIR synchronization, operators—including CNOs, MNOs, and MVNOs—exchange device-blocking lists through the GSMA CEIR and vendor-specific EIR interfaces. According to CEIR policies [38], an operator must block all devices contained in the downloaded list unless there is evidence of error or malicious intent. However, the CEIR interface defined in SG.18 provides only a minimal set of mandatory fields (e.g., record identifier. IMEI, status, action) in a device status record and lacks any provenance or security metadata (e.g., source of request). Critical information—such as reporter-identity verification status or device-ownership validation results—is entirely absent from the format. Vendor-specific EIR interfaces exhibit the same limitation.

Without such metadata, downstream operators have no basis for determining whether the contributor performed robust identity verification or ownership checks before submitting a block record. As a result, any operator with weak reporting safeguards can become an injection point for fraudulent entries—a problem we refer to as CEIR/EIR pollution. Once injected, these malicious records propagate across carriers during synchronization, enabling low-cost, cross-carrier device-blocking attacks. For example, an adversary can exploit a $5/month MVNO subscription (e.g., Tello) to submit fraudulent reports that ultimately affect higher-tier networks such as T-Mobile, dramatically reducing the cost of large-scale attacks.

Experimental Validation. As shown in §6.1, OP-I, OP-II, and OP-III do not participate in the GSMA CEIR program. We therefore evaluated the vulnerability that arises from EIR synchronization between MNOs and their MVNOs. For each operator, we selected two representative MVNOs based on market share (OP-Ia/OP-Ib, OP-IIa/OP-IIb, OP-IIIa/OP-IIIb) and conducted two tests: (1) reporting a phone as lost through an MNO and checking whether it remained usable on its MVNOs, and (2) reporting through an MVNO and checking usability on its host MNO and sibling MVNOs.

Figure 7 summarizes our findings. First, EIR pollution occurs in OP-I, OP-III, and their MVNOs. Adversaries can attack the MVNOs (OP-Ia, OP-Ib, OP-IIIa, OP-IIIb) by polluting their host MNOs' EIR databases, and can also attack the MNO (OP-III) by abusing the reporting service of one of its MVNOs (OP-IIIb). Second, EIR pollution does not occur in OP-II and its MVNOs. The tested MVNOs may maintain their own EIR databases, separate from their host MNO.

Root Cause and Lessons Learned. The root cause of this issue is often attributed to the shared EIR infrastructure between MNOs and their MVNOs, which allows attacker-induced pollution to propagate across operators. However, simply isolating EIRs is not a viable solution, as cross-operator synchronization is required by FS.45 for global device-blocking interoperability. The real vulnerability lies in the CEIR/EIR provisioning interfaces: operators receive block entries without the necessary information—such as reporter-identity verification status and device-ownership validation results—needed to distinguish legitimate reports from those submitted with malicious intent.

A key lesson is that CEIR/EIR exchanges must include mandatory security metadata to prevent malicious block records from propagating through the shared ecosystem.

7 Home Security System Freezing

We present secFreezing, an attack exploiting vulnerabilities V1–V4 to disable home security systems by blocking critical alarms from reaching homeowners and monitoring centers. A typical deployment includes a gateway coordinating wireless sensors (entry, glass-break) and maintaining continuous communication with a remote service team.

To ensure resilience against connectivity failures, most security gateways support dual networking: Wi-Fi as the primary channel and cellular as the fallback. Although both can be jammed [44, 69, 96], effective cellular jamming requires sustained high-power transmissions, which can be detected [21, 69] and localized [31, 67, 80] by operators using neighboring 5G/4G base stations, making it non-stealthy. In contrast, Wi-Fi jamming is more stealthy. Although it can be detected by analyzing signal features and connectivity patterns that deviate from normal interference, most consumer-grade devices lack this capability, as such detection requires specialized firmware to interpret PHY-layer noise or dedicated detection hardware [95]. Furthermore, even when Wi-Fi jamming is detectable, localizing the jamming source remains challenging due to the lack of coordination among neighboring Wi-Fi access points.

Because cellular links serve only as backup channels, gateways typically adopt low-cost cellular IoT technologies (LTE-M [41], NB-IoT [43]) for alerts or control messages, reducing hardware complexity (10–25% of LTE [86]), device cost, and data-plan cost [81].

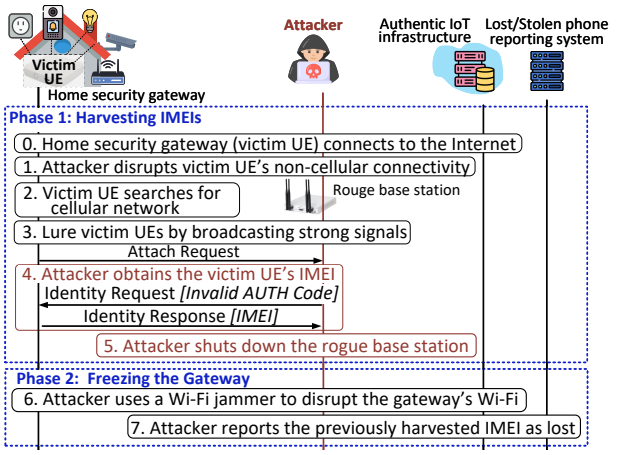


Figure 8: Overview of the secFreezing attack.

However, this design unexpectedly enables a far stealthier threat: a reporting-service–based freezing attack requiring no long-term jamming. It can fully block alarm delivery, causing financial loss to homeowners and major reputational damage to vendors in a market projected to reach USD 72.4 billion by 2025 [56]. Notably, in the U.S., roughly 75% of households deploy home security solutions, and 32% use alarm systems [72].

Adversary Assumptions. The adversary is assumed to: (1) identify households using home security systems, e.g., via visible yard signs; (2) disrupt the victim's Wi-Fi using Wi-Fi jammers [44, 69, 96] or deauthentication tools [33]; and (3) deploy an open-source SDR-based cellular IoT platform to interact with the victim's security gateway.

Attack Design. The secFreezing attack consists of two decoupled phases (Figure 8), allowing adversaries to harvest a gateway's IMEI at one time and trigger disruption weeks or months later—greatly increasing stealth and hindering forensic attribution.

◇*Phase I — Harvesting IMEI*[1]*:* The adversary briefly disrupts the gateway's Wi-Fi—e.g., using a Wi-Fi jammers or deauthentication tool [33]—forcing it onto cellular. A rogue base station with a stronger signal then lures the gateway and intercepts its signaling. Upon receiving the Attach request, the adversary sends a forged Identity Request exploiting V1, prompting the gateway to reveal its IMEI. The rogue base station is immediately shut down, leaving no persistent wireless footprint for operator detection.

◇*Phase II — Freezing the Gateway:* Weeks or months later—after any link to Phase I has faded—the adversary again disrupts the security gateway's Wi-Fi and reports the harvested IMEI as lost, thereby completely crippling the gateway's connectivity with the backend support team. As a result, intrusion or other critical alerts never reach homeowners or emergency responders. Notably, Phase II requires no sustained cellular jamming, making the attack practical and highly stealthy.

Attack Implementation. We purchased a home security gateway from HA-I, a representative U.S. vendor, subscribed to its remote monitoring service, and connected it to the Internet via a Wi-Fi router. We then built a rogue cellular IoT base station using Sonica [29] to transmit stronger signals and harvest the gateway's IMEI. To avoid unnecessary wireless interference, we adopted two approaches: (1) limiting the rogue base station's communication range to 10 meters, although it can be extended to tens of meters with an RF power amplifier [54]; and (2) emulating Wi-Fi jamming by manually disconnecting the Wi-Fi router's Internet cable, instead of deploying an actual jammer.

[1]To the best of our knowledge, there are no deployed or standardized authentication mechanisms which prevent adversaries from harvesting IMEIs from cellular IoT devices vulnerable to V1. As a mitigation, home security vendors can employ signal-strength–assisted authentication, using deviations in RSSI from historical profiles to detect rogue base stations and terminate connections to prevent IMEI exposure.

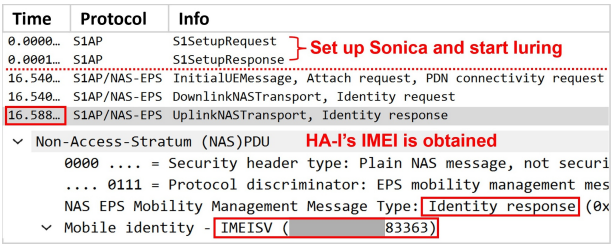
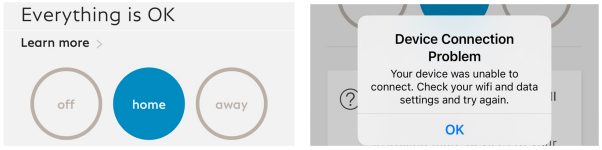


Figure 9: HA-I's IMEI is obtained within 17 seconds.



(a) Communication resumes first. (b) Communication freezes later.

Figure 10: Evaluation of home security system freezing attack against HA-I security gateway.

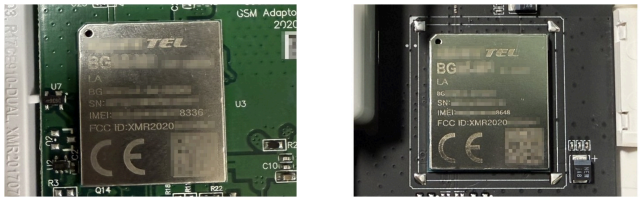


Figure 11: Two U.S. representative home alarm systems using the same vulnerable cellular IoT modem chipsets (Left: HA-I, Right: HA-II).

Attack Evaluation. We evaluate the attack across its two phases. We have two observations. First, in Phase I, the rogue cellular IoT base station harvests the gateway's IMEI within 17 seconds (Figure 9). Second, in Phase II, the HA-I gateway resumes cellular communication about 15 seconds after Wi-Fi disruption (Figure 10(a)). Once we report the gateway's IMEI as lost to OP-II (the operator collaborating with HA-I), remote access becomes unavailable, confirming the attack's feasibility (Figure 10(b)).

To assess broader impact, we also examine HA-II, another major U.S. provider. Both providers rely on vulnerable cellular IoT modem chipsets, as shown in Figure 11, suggesting the attack likely applies across the U.S. market. Notably, HA-I and HA-II hold over 41% of the U.S. market share.

Attack Implication. The real-world impact of this proposed attack extends beyond home security systems and applies broadly to devices equipped with cellular IoT modems. Many such devices operate in safety- or infrastructure-critical environments, including water and electricity meters [65], industrial sensors, and medical

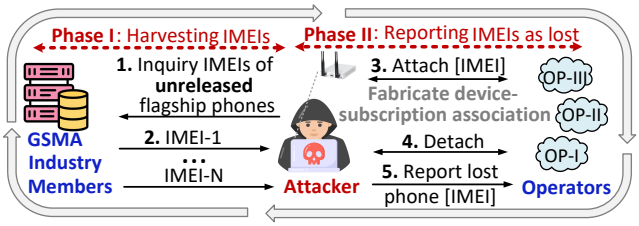


Figure 12: Overview of the flagAmbush attack.

monitoring systems such as Philips MCOT [68], Boston Scientific's BodyGuardian MINI PLUS [26], and Rhythmedix RhythmStar [71].

DoS attacks against these cellular IoT modems are especially concerning, as connectivity loss may compromise patient safety, delay medical intervention, or disrupt critical utilities and industrial operations. These risks show that secFreezing is not merely a localized issue but a systemic threat to the cellular IoT ecosystem, requiring urgent attention from manufacturers, operators, and standardization bodies.

8 Zero-Day Flagship Phone Ambush

We introduce flagAmbush, a proof-of-concept attack that exploits V2, V3, and V6 to preemptively blacklist unreleased flagship phones, e.g., Samsung Galaxy Z Fold 7 or Apple iPhone 17 Pro, across multiple carriers before these devices reach customers. In practice, early customer feedback strongly shapes a product's success. Severe launch issues—such as cellular connectivity failures—can harm public perception and even depress stock prices, as seen with the Samsung Galaxy Note 7 [52] and BlackBerry PlayBook [37].

Adversary Assumptions. The adversary (1) can anonymously access operators' self-reporting portals (via V2), and (2) can deploy an SDR-based UE to attach with a spoofed IMEI and fabricate a device–subscriber association (via V3).

Attack Design. It comprises two phases (Figure 12).

◇*Phase I — Harvesting IMEIs:* Adversaries can obtain IMEIs and the associated cellular carriers of unreleased flagship phones through GSMA industry members. Besides operators, GSMA grants entities like dotmobile's DCS and MobiCode access to its IMEI database for verification and supply-chain use. Yet GSMA provides no clear, enforceable protections for this data, allowing pre-market device information to be mishandled or leaked.

◇*Phase II — Reporting IMEIs as Lost Cross Carriers:* Adversaries use each carrier's self-reporting portal to mark devices as lost. Step 1: an attack UE attaches to the target carrier with a collected IMEI. Step 2: after receiving an IP address and waiting briefly (OP-I/II: 60 s; OP-III: 1 s), it detaches. Step 3: the reporting tool logs into the carrier portal and files a lost-device report, blacklisting the IMEI. Steps 1–3 repeat until all IMEIs are reported.

Attack Implementation. We obtained flagship-phone IMEIs from IMEI.info [47], a GSMA industry member ($600 for the full database, $30/month for updates). We purchased the service from IMEI.info and found that the IMEI of the $2,000 Samsung Galaxy Z Fold 7 was available before its 07-25-2025 release (Figure 13), confirming that adversaries can obtain device identifiers prior to launch. We built an SDR-based UE using srsRAN [77] on a laptop (Intel i7-1195G7) and developed a Python-based controller and reporting tool. Given a list of IMEIs, the controller instructs the SDR UE to spoof each IMEI and attach using the adversary's prepaid subscription. The reporting



Figure 13: The IMEI of a Samsung Galaxy Z Fold 7 was obtained from `IMEI.Info` prior to its official release.



(a) The IMEI of an unboxed Samsung Z Fold 7. (b) Device setup. (c) Blocked by OP-III.

Figure 14: A brand-new Samsung Galaxy Z Fold 7 is blocked after unboxing due to flagAmbush attack.

tool then uses Selenium [58] to automate carriers' self-reporting portals and submit lost-device reports for all spoofed IMEIs.

Attack Evaluation. Due to ethical and legal constraints, we cannot evaluate flagAmbush at scale. Instead, we assess two aspects: effectiveness—whether an adversary can report a brand-new flagship device as lost before first use—and scalability—the tool's ability to report multiple devices across carriers and the associated cost.

◇*Effectiveness:* We purchase a brand-new Samsung Galaxy Z Fold 7 (exclusive to OP-III) and obtain its IMEI without unboxing the device, as the IMEI is printed on the retail box (Figure 14(a)). We find that the retrieved IMEI is listed on `IMEI.info`, indicating that IMEIs of unopened devices are accessible through GSMA industry members, thereby paving the way for launching flagAmbush attacks. Next, we report the unopened Galaxy Z Fold 7 as lost using our tool prior to unboxing. We then unbox the phone and activate it with a new user account, emulating a typical launch-day buyer scenario. We observe that the device cannot connect to OP-III or its MVNOs (OP-IIIa, OP-IIIb), as shown in Figure 14, confirming the effectiveness of flagAmbush.

◇*Scalability:* We evaluate the tool's scalability using three metrics: (1) Device Reporting Time (DRT) — the time to report a device as lost, including associating it with the adversary's subscription; (2) Report Capacity per Account (RCA) — how many devices a prepaid account can report as lost; and (3) Per-Device Blocking Cost (PDBC) — the cost to block a device on a carrier.

Across 10 runs—each reporting a different IMEI—per carrier, OP-III shows the fastest DRT (19.39 s), while OP-I and OP-II are 60 s slower due to longer device–subscription association. All three carriers allow a prepaid account to report 10 devices as lost without errors, yielding RCA $\geq$ 10. The PDBCs at submission time are $2.5 (OP-I), $3.5 (OP-II), and $4.0 (OP-III). Thus, reporting 100 Samsung Galaxy Z Fold 7 devices would require 10 prepaid accounts ($250–$400) and 33–133 minutes, resulting in $200K worth of blocked devices.

Notably, as a proof of concept, we did not minimize cost or optimize tool throughput; adversaries could further lower costs via cheaper MVNO plans (see V6) and increase throughput using multithreading on the tool.

Attack Implication. The flagAmbush attack reveals systemic weaknesses in how pre-market device identifiers and lost-device reporting services are handled across the mobile ecosystem. A coordinated attack could block hundreds of new flagship phones from connecting on launch day, triggering customer complaints, negative reviews, and reputational and market damage. These risks underscore the pressing need for GSMA to enforce stronger data-protection rules for its members and to strengthen lost-device reporting services against abuse.

9 Solutions

Practical Challenges and Design Principles. Designing a fully secure solution for lost and stolen device reporting is challenging due to diverse deployment scenarios across carriers. For example, carriers may serve users without SSNs or government-issued IDs for business reasons. Excluding such users from reporting is impractical, as most are benign; however, relaxing requirements introduces exploitable attack surfaces.

Thus, our goal is to balance security and usability without increasing the reporting burden for most users. To this end, we (1) preserve existing workflows for the majority of legitimate users, (2) apply stronger verification only when necessary, and (3) provide alternative paths for users who cannot meet standard requirements. These approaches maintain inclusiveness across diverse use scenarios while raising the bar for attacks.

Proposed Solutions. We propose four solutions as follows:

- **S1: Extended IMEI Protection Conformance Testing (V1).** We propose extending the 3GPP UE conformance test suite to validate IMEI-handling behavior under all NAS Security Header Types. For each header value (0x0–0xF), the test harness sends forged Identity Request and Security Mode Command messages carrying an IMEISV Request IE with an invalid MAC or plain-text content. A UE passes only if it *never* returns IMEI/IMEISV under any unauthenticated or improperly protected NAS variant. This closes the standards–testing gap that enabled V1.
- **S2: Unified Reporter Identity Verification (V2).** We integrate carrier reporting portals with third-party ID-verification providers (e.g., ID.me [46], Stripe Identity [79]). These services already support U.S. federal agencies such as SSA and IRS, demonstrating national-scale deployability. Under this design, any subscriber must complete one-time government-ID verification before submitting online reports, while unverified users are redirected to in-store reporting.
- **S3: Multi-Dimensional Device–Reporter Verification (V3–V4).** We extend reporting portals with multi-dimensional validation that combines behavioral device–reporter association (via the 3GPP AF [10]) and subscription entitlement. Only devices that exhibit consistent usage and/or location relationships (e.g., call/attach patterns within the user's habitual activity area) *and* match the reporter's service plan are eligible for online reporting; others are directed to authenticated in-store verification. This addresses both usage-only weaknesses (V3) and missing entitlement enforcement (V4), yet backward-compatible with existing cellular architectures.
- **S4: Strengthened CEIR & EIR Participation and Security Metadata (V5–V6).** To prevent cross-operator exploitation of device blocking, we strengthen CEIR/EIR along two dimensions. First, CEIR/EIR participation should be mandatory—at least for tier-one carriers—to ensure propagation of legitimate block actions. Second, CEIR SG.18 and vendor-specific EIR provisioning formats must include mandatory security metadata, including reporter-ID verification status and device-ownership validation results. These fields allow receiving operators to identify dubious or malicious blocking records and prevent attacker-induced CEIR/EIR pollution.

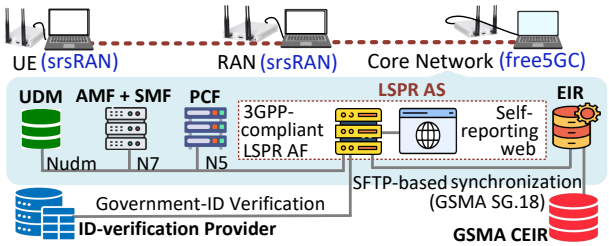


Figure 15: Illustration of solution prototype.

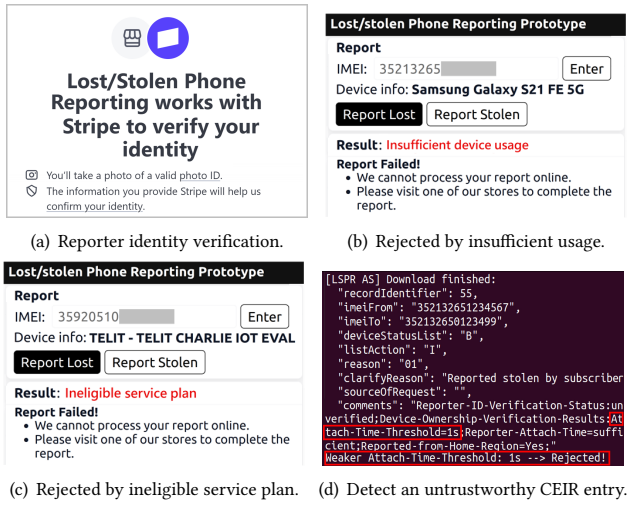
Impact on Reporting Usability and Scalability. Our design has minimal impact on usability and scalability. It preserves existing reporting workflows for most users, introducing additional verification only when necessary. Users who have already completed ID verification can report devices without extra steps, while others can perform a one-time verification via widely adopted, government-backed services (e.g., ID.me) before accessing the reporting system. Only high-risk reports are escalated to in-store verification. Moreover, our solution is compatible with standard 3GPP AFs and builds on the global CEIR architecture, which already supports hundreds of operators, thereby limiting manual handling to high-risk cases and enabling scalable deployment.

9.1 Prototype and Evaluation

Figure 15 shows our prototype built atop an SDR-based 5G testbed (srsUE [77], srsRAN [78], free5GC [35]). We implement an LSPR AS collocated with a 3GPP-stipulated Application Function (LSPR AF) and integrate it with a self-reporting web portal.

For S2, the portal uses Stripe Identity [79] to perform mandatory government-ID verification. For S3, the LSPR AS invokes the LSPR AF to gather device–reporter association metrics via standard interfaces (N5 [13], N7 [14], NUdm [12]), including registration duration, usage summaries, and subscription information. For S4, we implement a CEIR that supports SFTP-based synchronization and embed mandatory metadata—Reporter-ID-Verification-Status and Device-Ownership-Verification-Results—into the Comments field (textual comments) of a device status record [40].

The Device-Ownership-Verification-Results field uses compact carrier-threshold/criteria + result pairs, e.g.,: Call-Usage-Threshold = 10, Reporter-Call-Usage = sufficient; Data-Usage-Threshold = 100MB, Reporter-Data-Usage = sufficient; Attach-Time-Threshold = 10h, Reporter-Attach-Time = insufficient; Reported-from-Home-Region = Yes/No; Subscription-Entitlement-Criteria = none. These metadata enable downstream operators to assess block-entry credibility in a reporter-privacy-preserving manner while retaining flexibility to define their own verification criteria. For example, if the reporter is outside the device's habitual activity area, the carrier may treat the request as higher risk and restrict online reporting.



(a) Reporter identity verification. (b) Rejected by insufficient usage.

(c) Rejected by ineligible service plan. (d) Detect an untrustworthy CEIR entry.

Figure 16: Evaluation of the proposed solutions.

Evaluation Settings. For S2–S3, we configure three requirements for reporting eligibility: (1) registration duration ≥ 7 days, (2) data usage ≥ 100 MB, (3) subscription entitlement matching device type. We introduce two users: User-A (attacker), a new unverified subscriber with a non-IoT plan; and User-B (victim), an established user with smartphone and IoT plans. Devices IMEI-A and IMEI-B have never been used by User-A. User-A attempts to report both via the portal. For S4, we inject a malicious IMEI-block entry for IMEI-A into our CEIR instance, populating the Comments field with: Reporter-ID-Verification-Status = unverified; Attach-Time-Threshold = 1s; Reporter-Attach-Time = sufficient; Reported-from-Home-Region = Yes. The EIR then downloads the CEIR update to evaluate whether such metadata enables detection of attacker-generated entries.

Evaluation Results. Figure 16 summarizes our evaluation results. *S2:* Figure 16(a) shows that User-A is redirected for government-ID verification before submitting any report. *S3:* The LSPR AS rejects both reports due to missing historical usage and entitlement mismatches, as shown in Figure 16(b) and Figure 16(c), respectively. *S4:* The downloaded CEIR entry is classified as untrustworthy because its validation criteria fall below local thresholds; IMEI-A is therefore not blocked, as illustrated in Figure 16(d). This demonstrates a practical and privacy-preserving defense against CEIR/EIR pollution, consistent with GSMA FS.45, which permits operators to disregard CEIR-listed IMEIs when there is evidence of error or malicious intent.

10 Related Work

IMEI Spoofing and Defenses. Prior work shows that IMEIs can be spoofed on cellular IoT modules and SIM-box systems [61, 94], manipulated via SS7/MAP to bypass EIR controls [70], or abused to launch anonymous attacks on emergency services [45, 57]. Defenses attempt to detect device-side tampering [17, 30, 49, 51], use TEEs to protect identifiers [59], or validate IMEI consistency in the network [19, 34]. These approaches, however, require hardware support or modifications to 3GPP standards, limiting deployment across low-cost IoT devices and legacy networks. Carriers therefore continue relying on operational heuristics when mitigating IMEI spoofing against lost/stolen reporting services. Notably, prior work does *not* examine how IMEI spoofing interacts with the *end-to-end lost/stolen device reporting services*, including *CEIR/EIR propagation across operators*, nor does it identify the cross-carrier structural weaknesses we uncover.

Device-Blocking Architectures. Centralized CEIR-like registries [24, 36, 55] and decentralized blockchain-based IMEI databases [16, 62, 66, 73] aim to improve record integrity or transparency. Unlike these works, which propose new device-blocking architectures, we focus on the CEIR deployed by the GSMA, which has more than 1,000 members, including many tier-one operators. We examine the security weaknesses in the *existing* GSMA CEIR specifications and the carrier–CEIR/EIR synchronization procedures used in practice. We identify key issues that allow malicious block records to propagate across operators, and we propose deployable, backward-compatible remedies that strengthen global device-blocking effectiveness while preventing CEIR/EIR pollution attacks.

Device DoS Attacks. Prior DoS attacks rely on proximity via rogue base stations [25, 74, 83, 97] or sustained remote signaling abuse [53, 88, 91], typically degrading only specific services and being limited in range or stealth. In contrast, our reporting-based DoS requires no proximity, no continuous traffic, leaves no traceable reporter identity, disables the entire device, and propagates across carriers, giving it broader impact than the prior art.

11 Conclusion

In this study, we present the first end-to-end security analysis of lost and stolen phone reporting services across the device, carrier, and cross-carrier domains. Through real-world experiments on major U.S. carriers, we uncover six previously unknown vulnerabilities that enable stealthy and scalable DoS attacks. Using these flaws, we demonstrate two practical attacks, secFreezing and flagAmbush, which can disable cellular-based security infrastructure and preemptively block brand-new flagship phones at launch. These results show that a service designed for protection can instead be weaponized for large-scale disruption.

We further propose backward-compatible countermeasures and validate their effectiveness through a prototype implementation. Our findings expose a critical yet overlooked weakness in today's cellular ecosystem and call for coordinated efforts from carriers, vendors, and standards organizations to secure global device-blocking services.

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